

# PG-MDP: Profile-Guided Memory Dependence Prediction for Area-Constrained Cores

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## Abstract

Memory Dependence Prediction (MDP) is a speculative technique to determine which stores, if any, a given load will depend on. Area-constrained cores are increasingly relevant in various applications such as energy-efficient or edge systems, and can often only spare limited space for MDP tables. This leads to a high number of false dependencies as memory independent loads collide with unrelated predictor entries (i.e., the aliasing problem), causing unnecessary stalls in the processor pipeline.

This paper proposes a profile-guided hardware-software co-design to detect frequently memory independent loads. Our profiler labels those loads via their opcode to mark them as *predict-no-dependency* (PND) loads. PND loads bypass MDP and are always scheduled as early as possible, eliminating the possibility of false dependencies without introducing additional hardware overhead. Across SPEC CPU 2017 intspeerd, this technique achieves a geomean IPC gain of 1.4% and proportional power-savings in a small core configuration. On larger cores, performance is maintained, while average MDP power usage is reduced by 40%.

## 1 Introduction

Memory-level parallelism (MLP) is important for efficient out-of-order (OoO) execution and overcoming the memory wall in processor performance. Memory dependence prediction is a speculative technique used in OoO processors that aims to enhance total MLP by predicting the stores on which a given load instruction will depend, as well as identifying memory-independent loads. Smaller cores utilising smaller predictors can often struggle with a large number of false dependencies, which unnecessarily stalls loads on non-dependent stores. This is due to hash collisions in small predictor tables, or simple predictor algorithms that cannot capture varying dependency behaviour. The Store Sets predictor [?] exhibits both of these issues, and variations of the algorithm are found in real processors today [?]. Whereas modern predictors like PHAST [?] have pushed the SotA to address these issues via techniques like associative tagging and context based predictions, they place higher area, energy and complexity demands on the core design, which is often not viable in smaller cores. At the same time, while simply growing a Store Sets-like predictor would offset the hash collisions, this benefit struggles to outperform an iso-area comparison to scaling other components. [AJ: ref?](#) As such, memory dependence predictors in small cores typically remain small and inefficient.

Nevertheless, the importance of such area-constrained cores has grown over time. Modern systems increasingly deploy heterogeneous processors containing both performance and efficiency-oriented core designs, combining larger high-performance cores

with small energy-efficient (but still out-of-order) cores, as seen in ARM big.LITTLE systems and both Apple's & Intel's performance and efficiency cores. Efficiency-focused cores are, of course, also found in power-constrained systems such as smartphones, tablets, smartwatches and other edge systems. These cores are more and more performing demanding general purpose computation while striving to consume as little energy as possible. As they grow more performance-sensitive, finding optimisations which don't also incur additional area, power or complexity costs becomes more pressing.

This paper builds on an observation demonstrated by various prior works that a considerable portion of load instructions in general-purpose workloads rarely or never exhibit in-flight memory dependencies [? ? ?]. We leveraged this insight to reduce false dependencies in memory dependence prediction via a profile-guided software co-design at **zero hardware cost**. This profile-guided memory dependence prediction (PG-MDP) labels frequently memory-independent loads by their opcode, causing them to bypass MDP queries and always schedule as early as possible. In the event that a labelled load incorrectly passes a prior store to the same address, this still triggers a rollback as normal, but this will not create a new entry in the MDP. We show that on average 71% of dynamic loads can avoid querying the MDP while improving IPC by 1.4% on area-constrained cores.

## 1.1 Contributions

This paper makes the following contributions:

- Implements a profile-guided co-design technique to label memory independent loads to the OoO core at no hardware overhead.
- Implements a modified Gem5 to simulate labelled loads against different MDP algorithms and core size configurations.
- Extends McPat to provide power usage estimations of the MDP unit.
- Evaluates Spec2017 IntSpeed and demonstrates on average:
  - 71% of MDP load queries are redundant,
  - IPC improves on the smallest core by 0.4% and MDP power reduces by 12%.
  - the largest core maintains IPC and reduces MDP power by 42%.
- Additionally demonstrates a x% IPC gain when modelling a limited number of MDP read ports.

## 2 Background

This section outlines the fundamental concepts in memory-level parallelism for out-of-order execution. It then introduces Store Sets as a foundational memory dependence predictor algorithm, a modern variation found in the XiangShan processor, and lastly the current state of the art in the PHAST predictor.

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## 2.1 Loads in Out-of-Order Execution

The key structure in memory-level parallelism is the load-store queue (LSQ), used to hold in-flight memory operations and perform memory disambiguation. When load instructions dispatch, they are inserted into the load queue (LQ), and query the memory dependence predictor (MDP) for a predicted store(s) to wait on before execution. Once their source operands are ready and all predicted dependent stores have executed, loads search backwards through the store queue (SQ) for store-forwarding opportunities. If no matching addresses are found, the loads access cache to get their data.

When a load searches the SQ, not all earlier stores may have resolved their addresses. As the majority of the time the address will not match with the load, it is often profitable to speculatively ignore these stores and issue the load as soon as possible. When the unresolved store eventually receives its address, it searches forward through the LQ to find loads with matching addresses which have already executed. If a match is found, the load has read stale data and a memory order violation has occurred. The processor must flush all in-flight instructions from the load onwards and re-fetch.

The goal of the MDP is to minimise violation events while maximising the available MLP, allowing memory independent loads to issue as soon as possible and dependent loads to wait only for the necessary stores. When a load is incorrectly stalled on a store that does not resolve to the same address this is called a false dependency. Simple MDP algorithms typically prioritise reducing the number of violations over the number of false dependencies.

## 2.2 Store Sets

A seminal paper in memory dependence prediction is ‘Memory Dependence Prediction using Store Sets’ [?]. This is the MDP algorithm implemented in the open source Gem5 simulator [?]. Store Sets works by using Store Set IDs (SSID) to track dependent load-store pairs, assigning each instruction the same ID value in the PC-indexed Store Set ID Table (SSIT). The SSIDs are then used to index the Last-Fetched Store Table (LFST), which holds the sequence number of the most recently issued store with an SSID mapping to that LFST entry. Newly issued loads then access the LFST via their SSID to find the store they’re predicted to be dependent on. If a load PC does not map to a valid SSID entry, it is predicted to be memory independent. To forget old dependencies and reduce table pressure, the SSIT and LFST are cyclically cleared after a certain number of issued memory operations.

Store Sets is extremely area efficient, delivering a large portion of available performance with a very small hardware budget. Its small size and simplicity make it well suited to area-constrained processors. However, especially at smaller sizes, Store Sets struggles with false dependencies due hash collisions. When a memory independent load hashes to an existing SSID entry, it is incorrectly made dependent on the corresponding store for that entry. The clear period can be made shorter to offset this, but this comes at the trade-off of a higher number of memory order violations as the predictor must re-learn true dependencies more often. Furthermore, many loads exhibit non-consistent memory dependencies. A load in a loop may be dependent on a store every even iteration, but

not on odd iterations. Store Sets has no way to disambiguate these cases, and conservatively predicts the load as dependent on every iteration.

## 2.3 XiangShan Store Sets

XiangShan is an open-source high-performance RISC-V processor RTL implementation [?]. XiangShan represents the cutting edge in open-source superscalar processor design, and comes with a Gem5 fork modified to closer model the RTL implementation [?]. This Gem5 fork implements a variation of the Store Sets predictor that offers a closer representation to implementations found in real processors. From here on this is referred to as XS Store Sets.

The main optimisation over original Store Sets is using multiple ‘slots’ for each LFST entry, thereby recording multiple dependent stores at once. This allows a load to track multiple dependencies with only one SSID. This is useful in situations where a load is dependent on different stores in different program contexts, or may have partial address overlap with multiple stores. There are further small optimisations to allocation policy which we omit here.

## 2.4 PHAST

PHAST is a newer predictor which achieves much greater accuracy than Store Sets and other previously proposed MDP algorithms [?]. PHAST uses a TAGE-like structure to record a geometric series of branch history lengths between dependent load-store pairs. On dispatch, load instructions search all history length tables in parallel and selects the dependencies on the longest path (or predicts no dependency if no tables hit). This scheme addresses the previous problems in Store Sets by using multi-way tagged associative entries to reduce hash collisions, and predicting dependencies based on branch context to capture more elaborate dependency patterns. At a storage budget of 14.5KB, PHAST is shown to achieve within 1.5% accuracy of an ideal predictor.

Although PHAST delivers much greater accuracy, it also comes with greater hardware overhead. Though PHAST greatly outperforms Store Sets for iso-area comparisons at larger budgets, it struggles with the smaller budgets most efficiency-focused cores allow for memory dependence prediction. Furthermore, PHAST incurs pipeline complexity by requiring global branch history be made available in the load-store unit, has greater prediction latency, and consumes more power for the same area. This makes PHAST better suited only for larger performance-focused cores that would benefit most from the greater MLP.

## 3 Method

This section presents the profile-guided memory dependence prediction (PG-MDP) technique. We put forward the concepts of store distances, distance thresholds, and how these are combined to find reliably memory independent loads. We then motivate the use of profiles to determine load labels, outlining the benefits provided over static analysis.

[AJ: Before discussing the store distances, I suggest a subsection where Fig 1 is explained giving an overview of the technique. Reference the follow-up subsections for more details, but use this overview as the backbone of the method, to make it clear how everything fits together.](#)

### 3.1 Overview

?? I. Profiling step:

1. How is profiling performed? Binaries are compiled and run with different input tests **test** or **train**? provided with SPEC. For each load, one records the stores it depends on (store distance, see [subsection 3.2](#)).

2. Once the store distances are computed based on profiling, the info is recorded in the binary. How?

II. Execution:

The binaries are executed with the train ref input? How is the PND information retrieved by the hardware? How do the loads behave. See subsection X for details.

### 3.2 Store Distances & Labelling Thresholds

As overviewed in Section ??, when loads are issued in out-of-order execution they search the store queue (SQ) for older stores with overlapping addresses. We use the term store distance to refer to the number of prior entries in the SQ a load's dependent store is found. A load that depends on the most recent prior store would have a store distance of 1.

The profile-derived store distance for each load is found by recording every per-execution store distance on train inputs. Store distances that occur in 95% or more executions are selected as that load's profiled store distance. When no single store distance occurs in at least 95% of executions, the shortest observed distance is chosen. [AJ: How many inputs are used per binary? SPEC provides a low number of train inputs per benchmark, so it is not clear how many runs can be done and what means 95%.](#)

The premise our technique is based on is that loads with either no or consistently long dependent store distances are good candidates to label as PND. This heuristic captures three different types of behaviours:

- **(1) Memory Independent Loads:** Loads which read constant data (i.e. have no store distance).
- **(2) Rarely Dependent Loads:** Loads which observe short dependent store distances in <5% of executions.
- **(3) Structurally Independent Loads:** Loads whose dependent stores do not exist in-flight at the same time the load is issued.
- **(4) Fast-to-Resolve Dependencies:** Loads which have in-flight but resolved dependent stores, allowing forwarding.

[AJ: Can you connect the four categories with the profiling method: "The profile-derived store distance for each load is found by recording every per-execution store distance on train inputs. Store distances that occur in 95% or more executions are selected as that load's profiled store distance. When no single store distance occurs in at least 95% of executions, the shortest observed distance is chosen."?](#) [AJ: Maybe a code example with a CFG showing different store distances per branch and what is selected? Or several examples of increasing complexity?](#)

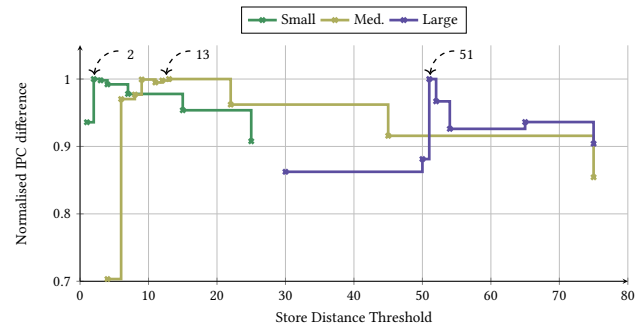
Loads with behaviour types one and two can always be safely labelled, but whether a load's store distance will fall into types three or four will depend on the target processor's characteristics. A larger processor with a longer SQ will hold more in-flight stores at once, so only loads with longer store distances will fall into this category. Type four is even further target (or even workload)

specific, but is still correlated with a longer distance as more time is left between dispatching the store and issuing the load. [AJ: The text above is correct, but I had to read it twice to follow the description :\) We can refine it in the final version.](#)

The optimal store distance threshold for a specific target processor is found via binary search, choosing the threshold with the best average performance over train inputs. [AJ: Threshold? We never talked about it before .. What is the optimal store distance threshold?](#)

By using a representative selection of workloads to perform the search with, an optimal store distance threshold can target the processor rather than each individual workload, meaning the search only has to be performed once. Figure 1 shows how the performance impact of labels for different core sizes varies, and the optimal choice for each target. [AJ: I suggest to move this figure and the discussion on the search of the optimal store distance threshold in a motivation section before the proposal description. Ideal store distance threshold is a good way to show the potential of the idea, once it was briefly presented in the intro. Then here present the method only, i.e. the implementation. Alternatively, the optimal store distance threshold can be considered as a pre-compilation step, i.e. same as profiling, so it can be presented there..](#)

Once a threshold is selected it is applied to each workload profile, and loads with sufficiently long distances have their opcodes changed to the appropriate PND load variant.



**Figure 1: Store Distance threshold search for each CPU model. The Arrows indicate the best threshold.**

### 3.3 Why Profiles?

While profile-guided optimisation (PGO) has long offered performance benefits, the technique has historically struggled to achieve adoption due to complicating the compilation workflow [?]. PGO also introduces concerns about the accuracy of generated profiles, and the potential to harm performance should real input differ too far from train input.

Addressing adoption, prior work shows that use of PGO has risen significantly in recent years [?]. This can be seen in enterprise projects such as Firefox and Chrome web browsers, the Python interpreter and the GCC compiler. Performance-critical server workloads have also seen renewed attention to the benefits that PGO can provide [?]. This enforces the use of profiles as a reasonable approach to designing new software-hardware co-designs.

We furthermore demonstrate in Section ?? that the our profiles generated from Spec2017 train inputs do generalise effectively to the test inputs, suggesting our proposed technique is able to generalise to unseen inputs. **AJ: From train to test is not "generalizing".. From train I would move to ref, or from test to train.. i.e. from smaller to larger inputs. It is expected that train "Generalizes" to test, because test is likely subsumed by train.**

It is also important to note that in the case of load instructions which are unseen in the train inputs, these are never labelled and so execute as normal. This allows our technique to be more tolerant to workloads with high code coverage variation between inputs.

Lastly we highlight the benefits that profiles are able to provide to this technique over static analysis. Of the four types of load behaviour referred to in Section 3.2, only two of them (types one and three) could theoretically be captured by perfect alias analysis. Type two would then further require perfect memory dependence analysis, and there exists no analysis in conventional compiler design that attempts to capture the behaviour in type four. It is also important to note that the alias and dependence analysis offered by industrial compilers today is far from perfect [? ]. Whereas stronger analyses do exist [? ? ], these are either only applicable to domain-specific languages or do not scale to large programs, making them either less applicable or less feasible. **AJ: In particular, I would highlight that most static alias analyses only work within the compile-time module, i.e. although they have support for inter-procedural analysis, they do not have whole-program visibility.** In contrast, profiles allow much greater flexibility and coverage of load behaviour, which pairs well with speculative execution where mistakes will not invalidate program semantics.

## 4 Evaluation

This section presents experimental results of applying PG-MDP to Spec2017 CPU intspeed workloads. It highlights how effectively false dependencies are eliminated and the resulting IPC gain on a small core (ROB=128), as well showing how the effects of PG-MDP scale to larger cores and state of the art MDP algorithms such as PHAST. This is followed by an analysis of how results change over different XS Store Set table sizes, and lastly provides an estimation of potential further performance gains when simulated a limited number of MDP read ports, limiting how many non-labelled loads can be dispatched per cycle.

### 4.1 Experimental Design

We use Gem5 [? ] to simulate varying core size configurations. The full configuration for each model is detailed in Table 1. We evaluate the Spec2017 Intspeed benchmarks using up to 10 sim-points, with an interval of 100M instructions and a warm-up of 10M. Gem5 is modified to bypass MDP lookups for labelled loads and to avoid creating new entries in the case of a memory order violation. Violating labelled loads still roll back as normal and do not alter program semantics. The Gem5 results are subsequently processed by an extended McPat, which models XS Store Sets or PHAST as applicable.

**Table 1: Parameters of processor components for each simulated model**

|                    |               | V. Small   | Small | Med.  | Large |
|--------------------|---------------|------------|-------|-------|-------|
| Instruction Window | ROB:          | 64         | 128   | 256   | 512   |
|                    | IQ:           | 52         | 77    | 154   | 308   |
|                    | LQ:           | 21         | 38    | 77    | 154   |
|                    | SQ:           | 12         | 22    | 54    | 108   |
| Pipeline Width     | Fetch/Commit: | 3          | 4     | 6     | 8     |
|                    | Issue:        | 6          | 8     | 8     | 12    |
| L1 Cache           | L1D (KB):     | 16         | 32    | 64    | 64    |
|                    | L1I (KB):     | 16         | 32    | 32    | 64    |
|                    | MSHRs:        | 4          | 8     | 16    | 32    |
| L2 Cache           | L2 (MB):      | 1          | 2     | 4     | 8     |
|                    | MSHRs:        | 8          | 16    | 32    | 64    |
| XS Store Sets      | SSIT:         | 32         | 64    | 128   | N/A   |
|                    | LFST:         | 16         | 32    | 64    |       |
|                    | Slots:        | 2          | 2     | 2     |       |
|                    | Clear:        | 125k       | 125k  | 125k  |       |
| PHAST              | Rows:         | N/A        | N/A   | N/A   | 128   |
|                    | Assoc:        |            |       |       | 4     |
|                    | Tag Bits:     |            |       |       | 16    |
| TAGE-SC-L Size:    |               | 64KB       |       | 128KB |       |
| Prefetcher:        |               | DCPT [ ? ] |       |       |       |

### 4.2 Performance

Figure 2 shows the percent changes in IPC across Spec2017 CPU intspeed for the small core configuration, achieving a 1.4% geomean improvement. This demonstrates PG-MDP’s ability to deliver zero-cost performance gains on area-constrained cores. It can be seen that IPC gains have an uneven spread across benchmarks, with some seeing large benefits and others next to none. This loosely correlates to the magnitude of false dependencies before and after seen in Figure 3. The four largest gainers - 625.x264\_s inputs 1 and 2, 641.leela\_s and 620.omnetpp\_s - all exhibit relatively higher false dependencies per kilo-instruction in the base case, and see this cut down significantly by over 80% by PG-MDP.

In select cases PND labels can lead to a performance regression. This is due to loads behaving memory independent on train inputs but exhibiting dependencies on ref inputs and causing an increase in memory order violations, as seen in Figure 4. Although memory order violations are increased in all workloads, they’re mostly offset by the reduction in false dependencies. It is worth noting that violations are already very rare events, measured in mega-instructions rather than kilo-instructions, so even a 100% increase can often still remain in negligible ranges compared to the number of false dependencies.

The degree to which violations increase is controlled by the threshold value discussed in Section ?. The optimal threshold may still provide a net performance gain while still causing small regressions in specific workloads, and a larger threshold could instead be chosen to ensure regressions never occur. They could also be mitigated by specialising the PGO technique to each benchmark rather than the processor, which is discussed further in Section ?.

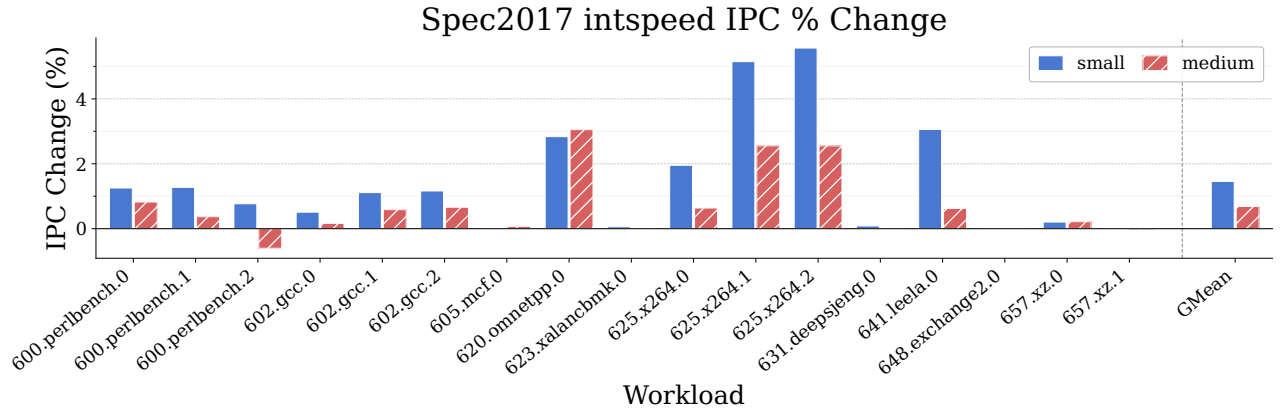


Figure 2: TODO: overview the important trend in each figure caption. IPC % changes for each CPU model on each intspeed benchmark.

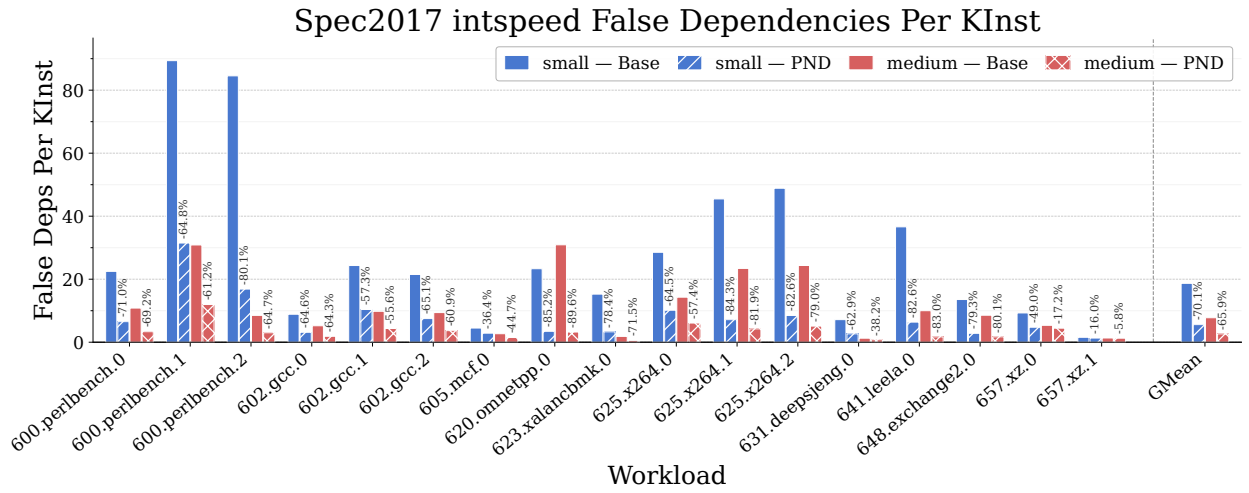


Figure 3: Percent change of false dependencies.

### 4.3 MDP & Core Size Sensitivity

Figure ?? plots a sweep of geomean performance in the base and PG-MDP case, for two cores and five XS Store Set sizes. This analyses how performance is impacted by core size and SSIT/LFST size. As expected, IPC gains achieved by PG-MDP fall sharply as SSIT/LFST sizes grow. This is because larger tables mean fewer hash collisions (and therefore fewer false dependencies) in the base case, and so less potential improvement in reducing them. The figure also demonstrates, again as expected, that the benefit of growing XS Store Sets on a small core quickly diminishes past small sizes, implying the additional area cost is better spent elsewhere. Table sizes become more important as instruction window length grows in the medium core, but still quickly reaches a plateau.

-talk about how labelled 64 matches base 128 on small and any other interesting points

### 4.4 Power Savings & PHAST

Figure ?? shows the total core energy reduction of each workload on the small configuration with PG-MDP applied. Because PG-MDP is a zero-cost optimisation, it can be seen that IPC gains directly translate into correlated power savings with no offset introduced by additional area costs.

Reducing MDP queries should in theory also offer a further power saving. However, we find that XS Store Sets already uses negligible power compared to the LSU and total core, that any reductions don't show up in final figures. Furthermore, XS Store Sets power usage is dominated by the sub-threshold leakage rather than runtime-dynamic cost (i.e. accessing the tables), and so an average 78% reduction in MDP queries only translates to an average 13% reduction in total MDP power.

However, the PHAST predictor incurs a greater power usage due to containing more tables and searching them all in parallel. This makes PHAST's runtime-dynamic power usage about equal

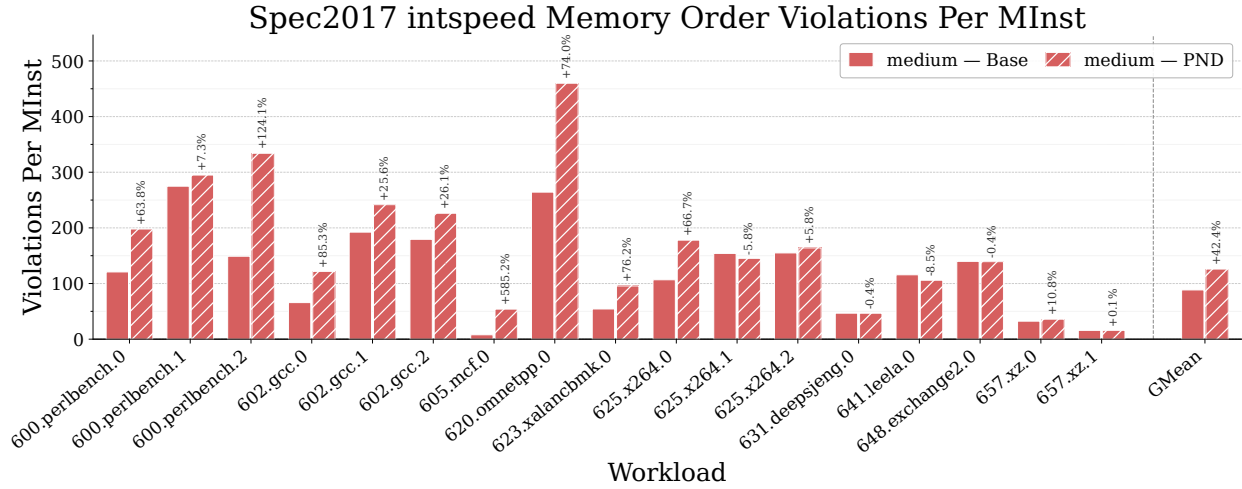


Figure 4: Percent change of memory order violations. Values roughly equal between core sizes.

to sub-threshold leakage, and takes up a larger proportion of total LSU power usage. This is especially true for larger cores that contain larger predictors. Evaluating PG-MDP on the large core configuration is shown to maintain performance while reducing average PHAST power usage by 40%.

- don't think we need to plot any graphs for the large core as they're not interesting, can just include the numbers - talk about total portion of LSU power, how its still quite small, but maybe future-proofs PHAST better

#### 4.5 Read Port Impact Estimation

Real processors have a limited number of read ports for each predictor component. If the memory dependence predictor has two ports, the processor can dispatch (that is, allocate instructions into OoO structures) up to two memory operations per cycle. Gem5 does not model this behaviour by default, as this is a low-level detail that is difficult to faithfully represent at the level of abstraction Gem5 simulates. Given the importance of this aspect to our proposed technique, we provide an estimation of the potential impact on IPC when the number of MDP read ports is limited. The dispatch stage is modelled to block when the number of unlabelled memory operations dispatched within a cycle exceeds the maximum number of read ports, which we model in the small core as 1. It is assumed that predictions always arrive within a single cycle, and so read port usage is reset to zero every cycle.

- insert results here

### 5 Related Work

*Improving Memory Dependence Prediction with Static Analysis* [?] tackles the same problem as this paper, but using static analysis to determine which load instructions to label as 'PND'. It demonstrates small IPC gains in select cases, but with an overall best-case gmean of less than 0.1%. Furthermore, due to relying on static analysis it also sees performance regressions elsewhere, making it unviable. In comparison, by leveraging profiles we are able to achieve higher IPC gains in broader contexts. Our technique is also resistant to

regressions on larger or more sophisticated MDP algorithms such as PHAST, which [?] is not.

*Feedback-directed Memory Disambiguation Through Store Distance Analysis* [?] proposes a similar profile-guided co-design to this paper, but with a more aggressive use case. In their work, the MDP is replaced altogether and load instructions are extended to include 4-bit distance fields indicating the store queue position they're likely to be dependent on. This achieves high accuracy compared to Store Sets, but comes at the cost of higher instruction bandwidth to encode predicted store distances, which is usually infeasible in modern processor designs where instruction bandwidth is critical. Larger processors would also likely require more bits to encode longer store queue distances. Furthermore, [?] could be more sensitive to profile accuracy than our work, as loads which do not appear in the profile must be assumed to be memory independent. In contrast, our technique is able to leave unseen loads to execute as normal.

*Software-hardware Cooperative Memory Disambiguation* [?] is a binary analysis co-design to reduce load queue pressure. Certain kinds of provably memory independent loads are labelled and prevented from being inserted into the load queue when issued. This technique is able to label 40% of static loads across Spec2006 floats (but dynamic percentage is unclear). This also incurs additional instruction bandwidth by inserting barrier-like instructions when entering loops to ensure labelled loads are only removed from the load queue when their parent loop reaches a steady state in the pipeline. LSQ entries must also store an additional bit to track whether these custom instructions are in-flight or not. This work presents an interesting proposal for reducing load queue pressure in area-constrained processors that lack space for hardware-based load elimination techniques, but has reduced viability due to currently lacking a mechanism for ensuring memory coherence in multi-core systems.

*Constable: Improving Performance and Power Efficiency by Safely Eliminating Load Instruction Execution* [?] is a pure hardware technique to track and eliminate stable loads which consistently read

the same value from memory. This technique is able to eliminate both the memory read and address calculation altogether, greatly improving pipeline efficiency. It also demonstrates a wide coverage a variety of workloads, improving gmean IPC by 5.1%. For larger performance-orientated processors, this technique would most likely be more effective at improving the efficiency of load execution than our work, with a large overlap (albeit not total) of the types of load instructions targeted. However, at an estimated hardware overhead of 12.4KB, this technique may be infeasible for the area-constrained cores we target.

*Other Memory Dependence Prediction Algorithms* Besides Store Sets and PHAST [??], other MDP algorithms include Store Vectors, NoSQ and MDP-TAGE [???]. Each of these algorithms offer different trade-offs and advantages. We choose to evaluate XS Store Sets and PHAST because we understand them to be the optimal choices for small and large cores respectively, and that XS Store Sets provides, to the best of our knowledge, the closest representation of MDP algorithms found in real processors.

## 6 Threats to Validity

This section touches on possible threats to the validity of results presented in this paper.

### 6.1 Encoding Space for Load Labels

We have proposed new opcodes to differentiate our labelled PND loads from regular loads in the ISA at zero overhead. While possible in theory, this may be difficult to implement in pre-existing and widely used ISAs such as X86, AArch64 and RISC-V due to limited available encoding space. Alternate proposals to this problem have been offered in [?]. Further options include introducing some level of overhead to apply our proposed technique, such as extending load opcodes to include a label bit, or reserving a bit in their register operand and adjusting register allocation accordingly.

### 6.2 Simulated Processor Models

The processor model we simulate specified in Section ?? does not reflect any real world processor configuration. Though we would like to use publicly known parameters from real processors (for instance from sources such as [?]), when using Gem5 to simulate workloads this can be dangerous, as the same features and mechanisms which inform the optimal parameter values in those real world processors could be absent in the Gem5 core. As such we design the simulated model used in this paper by the intuition of what is reasonable for Gem5 specifically, while still within realistic area bounds for a real processor in our use case.

### 6.3 Target Specific Compilation

The technique proposed in this paper assumed target specific compilation is possible in order to select the processor-specific optimal store distance threshold. In many cases though this is not the case. Enterprise software products are often compiled once and shipped to many different platforms. This would make short thresholds invalid, as they would introduce performance regressions in larger processors. A conservatively large threshold could be used for generic compilation, but this would likely also eliminate any

performance gains in small processors too. One solution could be for processors known not to benefit from PND labels to ignore the ISA hint and execute labelled loads as normal.

## 7 Future Work

*7.0.1 Per-Workload Threshold Selection.* As the threshold is untied to any architectural state, it is possible to select a different optimal store distance threshold for each individual workload. We chose not to do this for this work to prove that meaningful performance gains were possible with a generic threshold, and that the PG-MDP technique did not require a threshold search for each compiled program. However, there are use cases where this is feasible, and it would be worthwhile to evaluate the potential performance available for the additional compilation cost.

### 7.1 Just-in-Time Compilers

Just-in-Time Compilers (JITs) are a widely used run-time based compilation technique for dynamic languages. PG-MDP could potentially pair very well with these compilers for various reasons. Firstly, the additional workflow complexity of profiling can be automated by the run-time optimiser. Secondly, JITs make speculative optimisations about observed program behaviour, and so could potentially aggressively apply PND labels to any loads which read from addresses not recently stored to. Because PG-MDP is then also speculative, JITs do not necessarily have to immediately de-optimize if a labelled load does encounter a dependency. Lastly, dynamic languages tend to emit many memory independent pointer-chasing load instructions to implement data structures. This suggests these workloads could have a high potential for performance gains when using PG-MDP.

### 7.2 Serverless Workloads

Serverless workloads are characterised as a collection of many short-lived workloads which only execute a few times, then not again for a long time with lots of unrelated code in-between. This effectively makes their execution always cold on the target processor, meaning high performance is very difficult to achieve. When memory dependence predictors cannot learn dependencies fast enough to deliver performance gains (which is especially a concern for modern TAGE-like predictors such as PHAST), it may be preferable to disable memory dependence prediction entirely and conservatively stall loads on any unresolved stores. In this case, MD-PGO could potentially deliver substantial IPC gains by allowing likely-independent loads to issue immediately, while still preventing most memory order violations.

## 8 Conclusion

This paper proposed profile-guided memory dependence prediction (PG-MDP), a software-hardware co-design technique to label load instructions via their opcode as 'predict no dependency', and prevent them from making MDP queries to eliminate false dependencies. We showed PG-MDP very effectively reduces false memory dependencies at zero hardware cost, and delivers a geometric average of 1.4% IPC gain in a simulated area-constrained core, along with a correlated power saving. We further showed that this technique does not introduce regressions as processor size scales, and

813 can potentially offset the increased energy usage of modern MDP  
814 algorithms like PHAST in these processors.

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